

Discovering Latent Musical Structure Using Spotify Audio Features

Course: CS 4412 – Data Mining

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Key Exploratory Data Analysis (EDA) Insights

The dataset contains 114,000 Spotify tracks with 21 variables, including both metadata and numerical audio features such as danceability, energy, loudness, acousticness, instrumentalness, valence, tempo, duration, and popularity.

Data Quality Assessment

- Total observations: 114,000 Tracks
- Total Features: 21 variables
- Duplicate Rows: 0
- Missing Values: 3 total

The missing values occur in metadata fields (artists, album_name, and track_name). Importantly, no missing values are present in the numerical audio features used for clustering, so no imputation was required.

This confirms that the dataset is structurally clean and suitable for unsupervised learning.

Distributional Patterns

Exploratory visualizations (histograms, boxplots, and correlation heatmaps) revealed several key patterns:

- Energy and Loudness show strong positive correlation, reflecting production intensity.
- Acousticness is negatively correlated with energy and loudness, suggesting a production versus acoustic contrast.
- Popularity exhibits weak correlation with most audio features, indicating that commercial success is not strongly driven by any single musical dimension.

- Features such as danceability, energy, and valence are already scaled between 0 and 1, while duration (milliseconds) and tempo (BPM) have larger numeric ranges.

Overall, feature distribution indicate gradual variation rather than sharply separated categories. This suggests that musical styles likely exist along continuous dimensions rather than discrete classes.

Major Preprocessing Decisions and Rationale

Because K-Means clustering relies on Euclidean distance, preprocessing focused on ensuring fair distance calculations and meaningful feature representation.

Feature Selection

Only numerical features were retained for clustering. The following columns were removed:

- Unnamed: 0 (index column)
- Key, mode, and time_signature (integer-encoded categorical variables)

These categorical variables were excluded to avoid imposing artificial ordinal structure on nominal musical attributes.

The clustering model was built using the following numerical features:

- Popularity
- Duration (ms)
- Danceability
- Energy
- Loudness
- Speechiness
- Acousticness
- Instrumentalness
- Liveness
- Valence
- Tempo

Feature Scaling

Standardization (Z-score normalization) was applied using StandardScaler.

After Scaling:

- Each feature has mean ≈ 0
- Each feature has standard deviation ≈ 1

Scaling was necessary because duration and tempo have much larger numeric ranges than features bounded between 0 and 1. Without scaling, high-range variables would dominate Euclidean distance calculations and bias cluster formation.

First Mining Results and What They Mean

Clustering Method

K-Means clustering was implemented as the primary mining technique for M2.

Selecting the Number of Clusters

Two evaluation methods were used:

Elbow Method

The inertia curve shows a sharp decrease between $k = 1$ and $k = 3$, followed by a more gradual decline. This suggests diminishing returns beyond 3-4 clusters.

Silhouette Score

Silhouette values were:

- $K = 2 \rightarrow 0.1889$
- $K = 3 \rightarrow 0.0931$
- $K = 4 \rightarrow 0.0933$
- $K = 5 \rightarrow 0.0992$
- $K = 6 \rightarrow 0.0919$

Although $k = 2$ yielded the highest silhouette score, it produced overly coarse segmentation. $K = 3$ provided a reasonable balance between interpretability and structural separation. Therefore $k = 3$ was selected.

Cluster Characterization

Mean feature values and PCA projection revealed three interpretable musical archetypes.

Cluster 0 - Energetic / Dance-Oriented Tracks

- Energy ≈ 0.69
- Danceability ≈ 0.68
- Acousticness ≈ 0.27
- Tempo ≈ 118 BPM
- Popularity ≈ 32.9

This cluster represents rhythm-driven, production-heavy tracks optimized for movement and engagement. These songs likely correspond to energetic pop or performance-oriented music.

Cluster 1 – Acoustic / Low-Energy Tracks

- Acousticness ≈ 0.77
- Energy ≈ 0.28
- Loudness ≈ -14.77 dB
- Tempo ≈ 109 BPM
- Instrumentalness ≈ 0.32

These tracks are softer, more acoustic, and less-production intensive. The lower loudness and energy levels suggest emotionally subtle or stripped down musical compositions.

Cluster 2 – High Intensity / Fast Paced Tracks

- Energy ≈ 0.80
- Tempo ≈ 135 BPM
- Acousticness ≈ 0.09
- Danceability ≈ 0.48

This cluster captures high-tempo, high-intensity songs, potentially electronic, dance, or festival oriented tracks.

Overall Interpretation

This clustering primarily separates songs based on:

- Energy

- Acousticness
- Tempo
- Production Intensity

Importantly, clusters are not separated by popularity, reinforcing that commercial success is not the dominant structural dimension in this dataset.

The PCA visualization confirms:

- Visible but overlapping clusters
- Gradual transitions between musical styles
- Continuous rather than sharply separated musical structure.

This aligns with the modest silhouette scores, indicating that musical attributes vary along smooth latent dimensions.

Plan for M3

The M1 proposal included K-Means, DBSCAN, and anomaly detection.

Based on M2 findings:

Confirmed:

K-Means successfully identifies interpretable musical archetypes driven by energy, acousticness, and tempo.

Planned for M3

1. Implement DBSCAN to detect non-spherical cluster structures.
2. Compare density-based clustering results with K-Means.
3. Performs anomaly detection to identify musically extreme or unusual tracks.
4. Evaluate clustering robustness using additional validation metrics.
5. Explore whether clusters align with genre labels (track_genre)

Conclusion

K-Means clustering identifies three significant musical archetypes based on intensity, acoustic characteristics, and pace rather than popularity using 114,000 Spotify tracks and 11 standardized numerical audio attributes.

The results show that Spotify audio attributes encode interpretable latent musical structure, despite the moderate cluster separation.

These results provide a solid basis for more in-depth density-based grouping and anomaly investigation in M3.